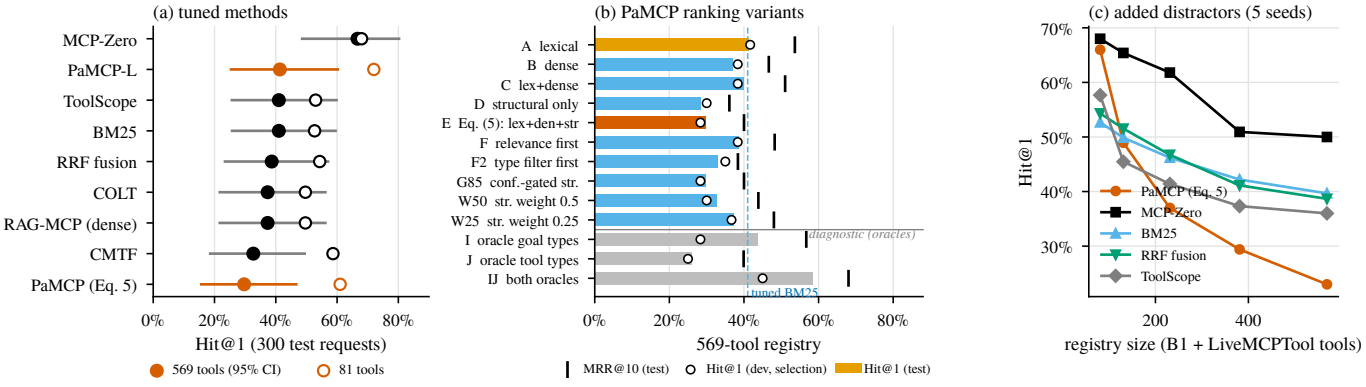
(a) tuned methods
MCP-Zero
PaMCP-L
ToolScope
BM25
RRF fusion
COLT
RAG-MCP (dense)
CMTF
PaMCP (Eq. 5)
0%
20%
40%
60%
80%
Hit@1 (300 test requests)
569 tools (95% CI)
81 tools
(b) PaMCP ranking variants
A lexical
B dense
C lex+dense
D structural only
E Eq. (5): lex+den+str
F relevance first
F2 type filter first
G85 conf.-gated str.
W50 str. weight 0.5
W25 str. weight 0.25
I oracle goal types
J oracle tool types
IJ both oracles
diagnostic (oracles)
tuned BM25
0%
20%
40%
60%
80%
569-tool registry
MRR@10 (test)
Hit@1 (dev, selection)
Hit@1 (test)
(c) added distractors (5 seeds)
70%
60%
50%
40%
30%
Hit@1
PaMCP (Eq. 5)
MCP-Zero
BM25
RRF fusion
ToolScope
200
400
registry size (B1 + LiveMCPTool tools)